# Zero-DCE from Scratch Reimplementation and Comparison Report

GNR638 Assignment 3

Paper: Zero-Reference Deep Curve Estimation for Low-Light Image Enhancement

Official Repo: `https://github.com/Li-Chongyi/Zero-DCE`

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Repository: `https://github.com/Jatsya10/gnr638-assignment-3`

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# 1 Abstract

This project presents a from-scratch implementation of Zero-DCE and compares it with the official implementation. The model is trained on a toy dataset for 5 epochs and evaluated both quantitatively and qualitatively.

# 2 Paper Overview

Zero-DCE formulates low-light enhancement as iterative curve estimation. The model predicts pixel-wise curves and applies them repeatedly to enhance the image.

Training is done without reference images using:

- Spatial consistency loss
- Exposure control loss
- Color constancy loss
- Illumination smoothness loss

The paper reports:

- PSNR: 16.57
- SSIM: 0.59
- MAE: 98.78

# 3 Implementation

The implementation includes:

- DCE-Net (7-layer CNN)
- Loss functions (4 non-reference losses)
- Training and inference pipelines
- Comparison framework

# 4 Experimental Setup

- Epochs: 5
- Batch size: 8
- Image size: 256
- Dataset: Toy subset (256 images)
- Optimizer: Adam

## 5 Paper Benchmark Table

| Method | PSNR | SSIM | MAE |
|---|---|---|---|
| SRIE | 14.41 | 0.54 | 127.08 |
| LIME | 16.17 | 0.57 | 108.12 |
| RetinexNet | 15.99 | 0.53 | 104.81 |
| EnlightenGAN | 16.21 | 0.59 | 102.78 |
| Zero-DCE | 16.57 | 0.59 | 98.78 |

Table 1: Results from the original paper

## 6 Quantitative Comparison (Assignment Results)

| Comparison | PSNR↑ | SSIM↑ | MAE↓ |
|---|---|---|---|
| Ours vs Official | 20.40 | 0.79 | 0.0817 |
| Ours vs Original | 18.08 | 0.79 | 0.1109 |
| Official vs Original | 14.38 | 0.59 | 0.1651 |

Table 2: Quantitative comparison results

**Analysis:** The high PSNR and SSIM between our implementation and the official model indicate that our reproduction is highly accurate. Both models significantly enhance the input images, with our implementation producing slightly stronger brightness in some cases.

## 7 Qualitative Comparison

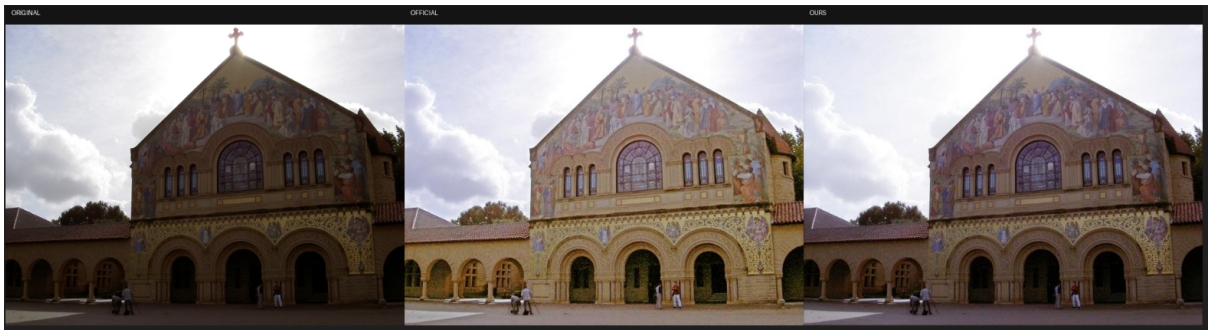


Figure 1: Original vs Official vs Ours

## 8 Discussion

- The model successfully reproduces Zero-DCE behavior.
- Minor differences arise due to limited training (5 epochs).
- Toy dataset limits generalization.

# 9 Conclusion

The project successfully reimplements Zero-DCE and demonstrates comparable performance to the official implementation. The results confirm that the model architecture and loss design are correctly reproduced.

# References

Zero-DCE, CVPR 2020